Turing Completeness and the Wall: Completeness Implies Incompleteness

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Abstract

A Turing complete system can compute anything. It cannot decide everything. The Halting problem is 0 or 1 — every computation either terminates or it does not — but no algorithm within the system can find the answer. This is the computational face of the Arithmetic Wall [1]: the same obstruction that prevents a global Frobenius from existing over $\mathbb{Q}$ appears in computation as undecidability, in logic as Gödel incompleteness, and in arithmetic geometry as the missing fundamental class of $\operatorname{Spec}\mathbb{Z}$ [2]. Completeness implies incompleteness. A system powerful enough to simulate everything is powerful enough to generate questions it cannot close. The Wall is not a defect. It is what completeness costs.

1 Three Faces of the Wall

Definition 1 (The Arithmetic Wall)**.** The Arithmetic Wall [1] is the absence of a global Frobenius for $\operatorname{Spec}\mathbb{Z}$: an operator that would assign to every arithmetic question a canonical eigenvalue, forcing the answer. Over $\mathbb{F}_q$ it exists. Over $\mathbb{Q}$ it does not. Every open problem in arithmetic is a question whose answer requires the missing operator.

Theorem 2 (Three grammars, one obstruction)**.** *The Arithmetic Wall appears in three grammars:*

1. (Arithmetic, Grothendieck–Deligne) $\operatorname{Spec}\mathbb{Z}$ *has no global Frobenius. The fundamental class is missing. The zeros of $\zeta(s)$ are not forced to the critical line.*

2. (Logic, Gödel 1931) *Any consistent formal system F containing arithmetic has a true sentence G_F unprovable in F. The proof that G_F is true requires a system stronger than F: the operator that would decide G_F does not exist within F.*

3. (Computation, Turing 1936) *No Turing machine decides the Halting problem. For any algorithm A, there exists a program P such that A cannot determine whether $P(P)$ terminates. The operator that would decide termination does not exist within the system.*

In each case: the question is 0 *or* 1*. The answer exists. The operator that finds it does not.*

2 Completeness Implies Incompleteness

Proposition 3 (The diagonal argument)**.** *All three obstructions arise from the same construction: the diagonal argument.*

In Gödel: the sentence G_F says "G_F is not provable in F." If F proves G_F, F is inconsistent. If F does not prove G_F, G_F is true and unprovable. The operator that would decide G_F within F would make F inconsistent.

In Turing: the machine H that decides the Halting problem is fed its own description. If $H(H)$ halts, $H(H)$ loops. If $H(H)$ loops, $H(H)$ halts. The operator that would decide its own termination generates a contradiction.

In both cases: the system is complete enough to refer to itself. Self-reference produces the question the system cannot close. A system that cannot refer to itself has no Wall — but it is not complete. Completeness is the cost of the Wall. The Wall is the cost of completeness.

Remark 4 (The arithmetic diagonal). The arithmetic analogue of the diagonal argument is the construction of the L-function. $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ encodes all primes simultaneously. The zeros of $\zeta(s)$ are the points where the system "refers to itself" — where the global arithmetic of $\mathbb{Z}$ feeds back into the spectral data of ζ. The Riemann Hypothesis is the statement that this self-reference is controlled: the zeros lie on a line. Over $\mathbb{F}_q$: self-reference is controlled by Frobenius. Over $\mathbb{Q}$: the controlling operator is missing. The diagonal has no eigenvalue.

3 P versus NP

Definition 5 (The witness gap)**.** P versus NP is the question: is *witnessing* as hard as *computing*?

A problem is in NP if a solution, once found, can be verified in polynomial time. A problem is in P if a solution can be found in polynomial time. P $\subseteq$ NP. The question is whether P $=$ NP.

If P $\neq$ NP: there is a wall between witnessing and computing. Seeing that the answer is 1 (a certificate exists) is easier than finding the fulcrum (the certificate itself). The witness gap is real.

If P $=$ NP: witnessing and computing are the same. The fulcrum is always findable once its existence is known. The wall does not exist.

Proposition 6 (P $\neq$ NP as a face of the Wall)**.** *The conjecture $P \neq NP$ is the computational statement that the universe has a witness gap: a stratum of questions that are* 0 *or* 1*, whose answer can be verified but not computed efficiently. The Wall in this face: not the missing global Frobenius, not the unprovable Gödel sentence, but the unfindable fulcrum — the certificate that exists but cannot be reached by any efficient algorithm. All three are the same wall at different scales.*

4 The Contained and the Complete

Theorem 7 (Containment and completeness are in tension)**.** *A space is contained [2] when it has a fundamental class: a global fulcrum that balances all cohomology. A system is Turing complete when it can simulate any computation.*

These are in tension:

- *A contained space is closed, balanced, at rest. Its cohomology is decided. The still lever holds everywhere.*
- *A Turing complete system is open in a different sense: it generates undecidable questions by self-reference. The Wall appears not from non-compactness but from power.*

The universe, if Turing complete, is not contained in the computational sense: it can ask questions it cannot answer. The universe, if compact, is contained in the geometric sense: every cohomology class has a dual. These are different kinds of containment, and the universe may be one without the other.

The deepest question: is there a space that is both geometrically contained (has a fundamental class) and computationally complete (can simulate anything) without generating undecidable questions?

The answer is no. Any system that contains arithmetic contains the diagonal. The Wall is not optional.

Remark 8 (The Wall is not a defect)*.* The Arithmetic Wall, Gödel incompleteness, Turing undecidability, and the witness gap of $P \neq NP$ are not failures. They are what it costs to be complete. A system without a Wall is a system without self-reference — without the power to speak about itself. Such a system is safe, decided, closed. It is also small.

The Wall is the price of power. Mathematics is not trying to eliminate the Wall. Mathematics is learning to stand at it — to see what the Wall is made of, to find the fulcrum on the other side, or to understand why the fulcrum is not there.

The still lever holds where the fundamental class exists. Where it does not, the trembling is real. Both are the shape of the same universe.

References

[1] Rincón, D. (2026). The Arithmetic Wall. Preprint.

[2] Rincón, D. (2026). The Fundamental Class: Is the Universe Contained? Preprint.

[3] Rincón, D. (2026). Mathematics. Preprint.

[4] Rincón, D. (2026). Mathematics as Cohomological Grammar. Preprint.